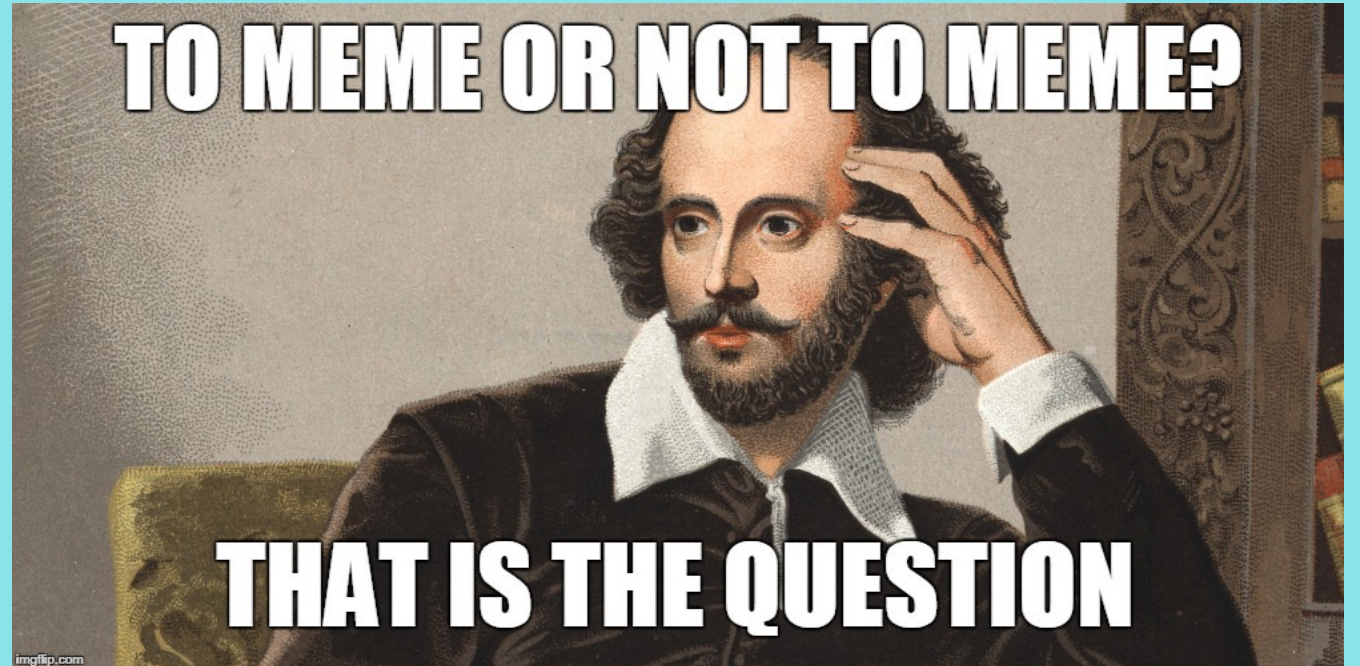


Memes



<https://www.govloop.com/community/blog/6-things-to-consider-before-sharing-a-meme/>

George, Lu, Deanna, Inga, Milos

SICSS Norrköping , 19-06-2025

Introduction

Meme is defined as a unit of cultural information, phrase, image, or video, that is transmitted from one individual to another (Das, 2023).

Denisova (2019) is going even further, calling them alternative media vehicles and “political mind bombs”.

The term "meme" was originally introduced by Richard Dawkins (1976) to denote an idea, behaviour, or style that spreads from person to person within a culture.

Background

Mememes are not just a trivial phenomenon of digital culture (Blackmore, 2000).

Mememes play a pivotal role in shaping online discourse, influencing cultural trends, and even driving public opinion (Das,2023; Sah & Jordan, 2023).

Nieubuert (2021) argues that mememes are like modern-day leaflet propaganda, that exploits cognitive biases (confirmation bias, negativity bias, humor, bandwagon effect).

Mememes can unite people through shared humour or divide them through misinformation i.e., contribute to social cohesion or division.

Theoretical Framework

Memetics theoretical framework provides a macro-level evolutionary model of how memes spread and survive like cultural genes (Dawkins,1976).

Memes are mirroring the process of natural selection, extending Darwinian idea of evolution beyond biology.

Social Cognitive Theory (Bandura, 1986) offers a useful framework for understanding the dynamic of memes from an individual perspective, through the process called observational learning.

Users observe what other users share, and they learn that certain memes get attention (likes and comments), using them as measures of social approval.

Research Question

RQ1: Do memes impact post engagement?

RQ2: Do memes affect the type/sentiment of the discussion?

RQ3: How do different sub-communities engage with memes?

Data

We gathered data from following Reddit communities (subreddits) that *contain* both memes and non-meme content:

- r/datascience
- r/pcmasterrace
- r/ChatGPT

These subreddits were selected due to their dynamic meme-sharing cultures.

While our initial plan was to collect data for a year across all three communities, some attrition occurred over time.

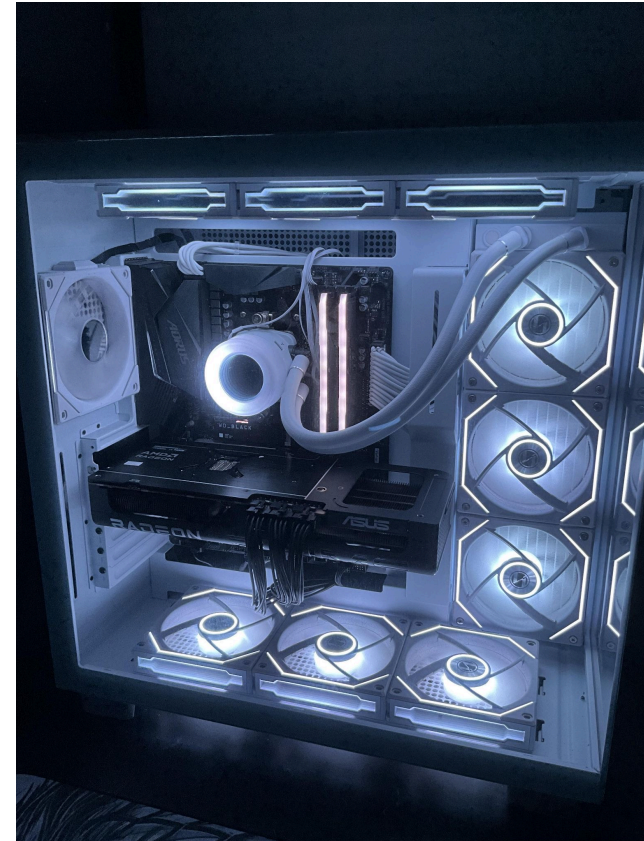
Despite these limitations, we believe that our dataset remains relevant for analyzing meme dynamics within these online communities.



Method (Overview)

1. Meme Detection (Training of two Models)
2. Sentiment Analysis of the Comment Section for each Post
3. Network Analysis

Meme vs Non-Meme



Meme vs Non-Meme

reddit

DS r/datascience Search in r/datascience

Home Popular Answers BETA Explore All

CUSTOM FEEDS

Create a custom feed

RECENT

COMMUNITIES

RESOURCES

About Reddit Advertise Reddit Pro BETA Help Blog Careers Press Communities Best of Reddit Topics

DS r/datascience • 20 days ago

Perfect job for me suffering from Imposter Syndrome

Career | Europe

16:11

Starling Bank

Data Scientist (Fraud)

Manchester, England, United Kingdom · 1 day ago · 92 applicants

Promoted by hirer · No response insights available yet

✓ Hybrid ✓ Full-time

Easy Apply Save

Use AI to assess how you fit

Get AI-powered advice on this job and more exclusive

1.7K 54 Share

sub-samples by n_size and % purity, color as % purity

Data Scientist / MLOps Engineer

Data Scientist with 6 years of experience in Supply Chain, Manufacturing, and Financial Analytics, specializing in Predictive Analytics, Early Warning Systems, and Spend Forecasting. Passionate about building scalable, data-driven solutions with AI that drive business impact, reduce cost & operations, and enhance decision-making.

SKILLS

Languages Python, SQL, PySpark, R | **Top Libraries:** pandas, numpy, Tensorflow, keras, sci-kit

Tech Stack Azure Tools, Databricks, Git, Spark, PowerBI

Technical Skills Machine Learning (Supervised & Unsupervised), Deep Learning, Predictive Maintenance, Anomaly Detection, LLM, Forecasting, Statistics, Time Series Analysis, Feature Engineering, Data Visualization

Domain Expertise Supply Chain, Healthcare, Manufacturing, Finance, Procurement

TECHNICAL EXPERIENCE

Data Scientist **Sep 2023 – Present**
Magna International

- Improved a legacy Anomaly Detection solution using OLS with sensitivity filters, reducing execution time from 6hr to 15min with Databricks Spark and MLflow. Automated triggers provide real-time updates, with an interactive chatbot, enabling informed budget planning and proactive actions on financial KPIs to C level executives.
- Led the development of a Global Warranty Early Warning System, unifying parts data from 15 OEMs for a comprehensive dataset. Used forecasting models to trigger alerts when predefined thresholds exceeded based on historical claims. Integrated the solution into a web app with developers and database teams, reducing operational and unexpected costs for the warranty team.
- Developed and deployed a Spend Forecasting model using FB Prophet, achieving 90% accuracy (MAPE of 0.1) for monthly spend predictions. Scaled the solution across the company globally, enabling procurement teams to optimize budget planning and negotiate better pricing with suppliers, resulting in millions of USD in cost savings.

Data Science Analyst **Sep 2021 – Sep 2023**
Accenture Strategy & Consulting

- Designed and deployed a scalable Anomaly Detection Model for a Fortune 500 Pharmaceutical Company, providing 95% accurate early failure warnings for critical manufacturing assets. Reduced unplanned downtime, improved asset life and reliability. Identified key failure drivers from sensor data and FFT Analysis, demonstrating their impact on asset life-cycle and leading to improved preventive maintenance.
- Led data collection and analysis for Predictive Maintenance feasibility studies, integrating sensor data, maintenance history, and cost estimates to assess ROI. This helped the executives on Go/No-Go decisions for building predictive solutions.
- Migrated legacy predictive models to the cloud, transforming them into plug-and-play accelerators that reduced project timelines and improved scalability for future engagements.
- Profiled quality data (audits, inspections, complaints, and deviations) from manufacturing sites to prepare a score matrix, forecast, and rank high-risk locations, helping leadership proactively mitigate operational threats.

Junior Decision Scientist **Oct 2019 – Sep 2021**
Mu Sigma

- Developed survival analysis models using Kaplan-Meier plots and Cox Proportional Hazards, identifying key survival patterns and improving stakeholder decision-making. Enhanced prediction accuracy by 15% using a WTE-RNN framework over traditional Cox models.
- Built an XGBoost-based classification model for propensity scores, increasing accuracy by 15% compared to logistic regression. Designed a feature selection framework to identify 30 important features from over 200, reducing training time by 35% while improving accuracy by 15%.

EDUCATION

Bachelor of Technology 2019
XII in Science 2015

ACHIEVEMENTS

12th Agis Graham Bell Awards for Innovation in Manufacturing (Accenture) 2022
STAR Award for Innovation and the Beacon Award from a Fortune 500 client, for excellence in data-driven solutions, client value creation, and leadership (Accenture) 2021
Spot Award by Fortune 500 Client (Mu Sigma) 2020

/* Don't worry, will optimize this query later */ DBA: "Oh, he must just be in a hurry to pull this data"

ONE YEAR LATER

DBA checking logs and seeing the same comment

datagoats.org

Image Classification Model

Model 1: Pretrained VGG19 (Image-Based)

- Convolutional Neural Network finetuned on meme data

Model 2: Llama 3.2 Vision (Image + Text)

- A vision-language model that can process both images and text (post title and content)
- Prompt:

`"Is this image a meme? The end of the response should contain a clear yes or no. The post title is: '{post_title}' and the image caption is: '{image_caption}'."`

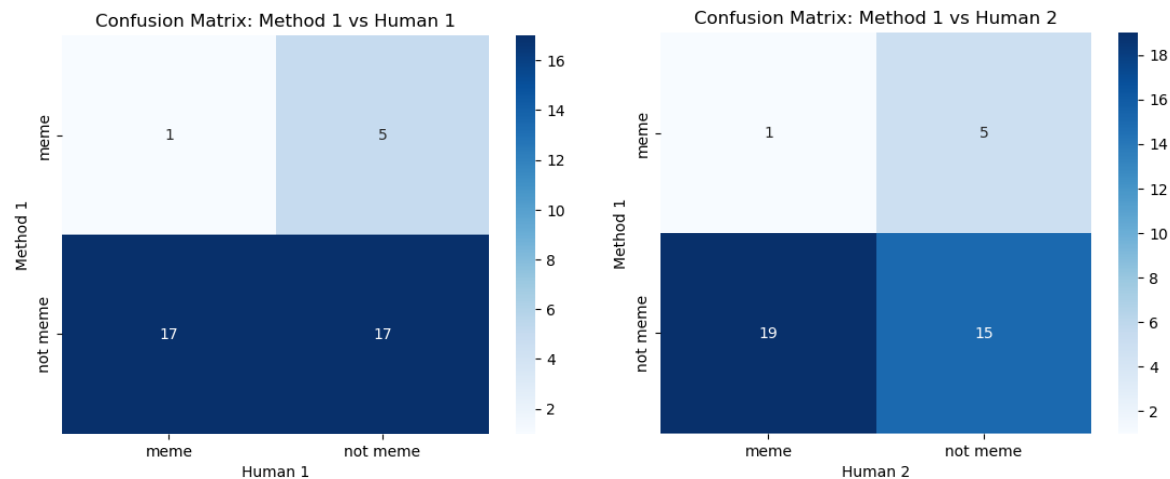
- Modified Prompt for indecisive answers:

`"Is this image a meme? Only respond with yes or no. You must answer yes or no. The post title is: '{post_title}' and the image caption is: '{image_caption}'."`

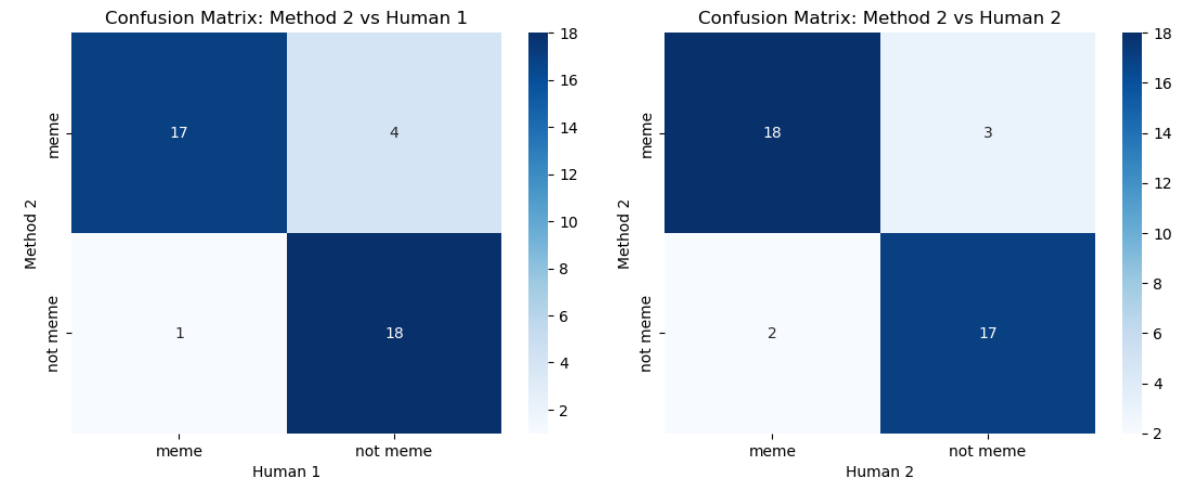
Classification Model Evaluation

- Two team members labeled images from r/datascience
- Model that performed best compared to human annotation was used

Model 1: Pretrained VGG19 (Image Based)

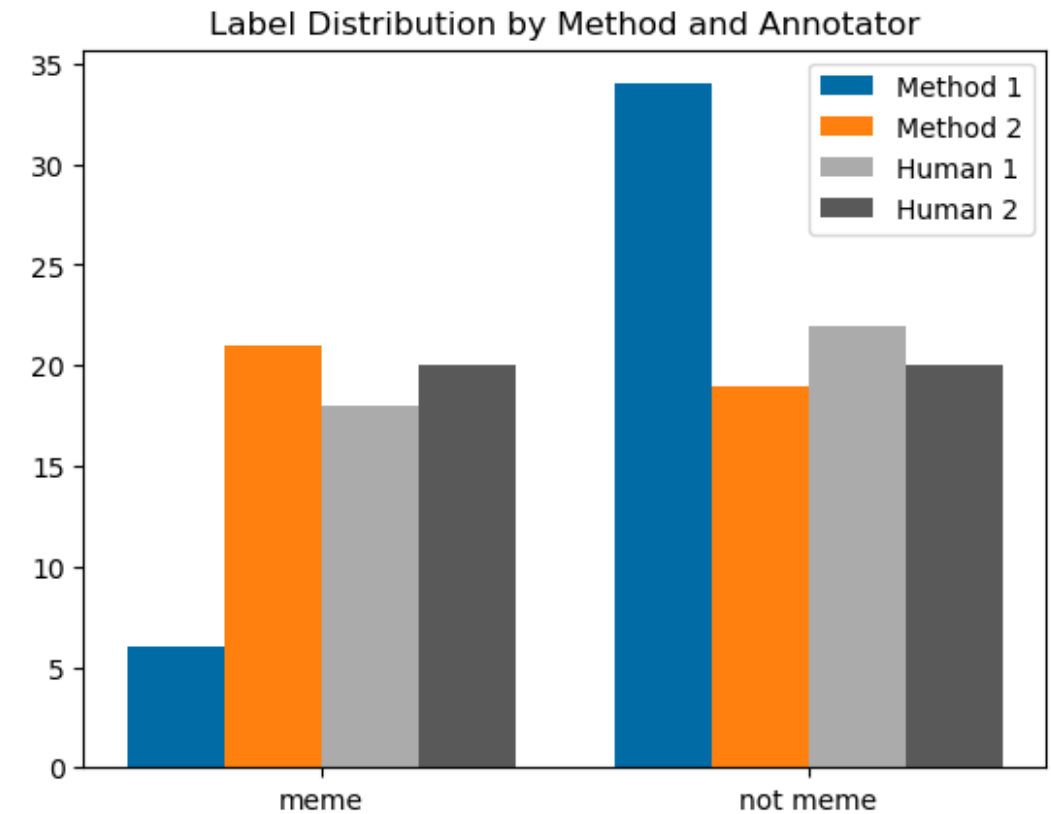


Model 2: Llama 3.2 Vision (Image + Text)

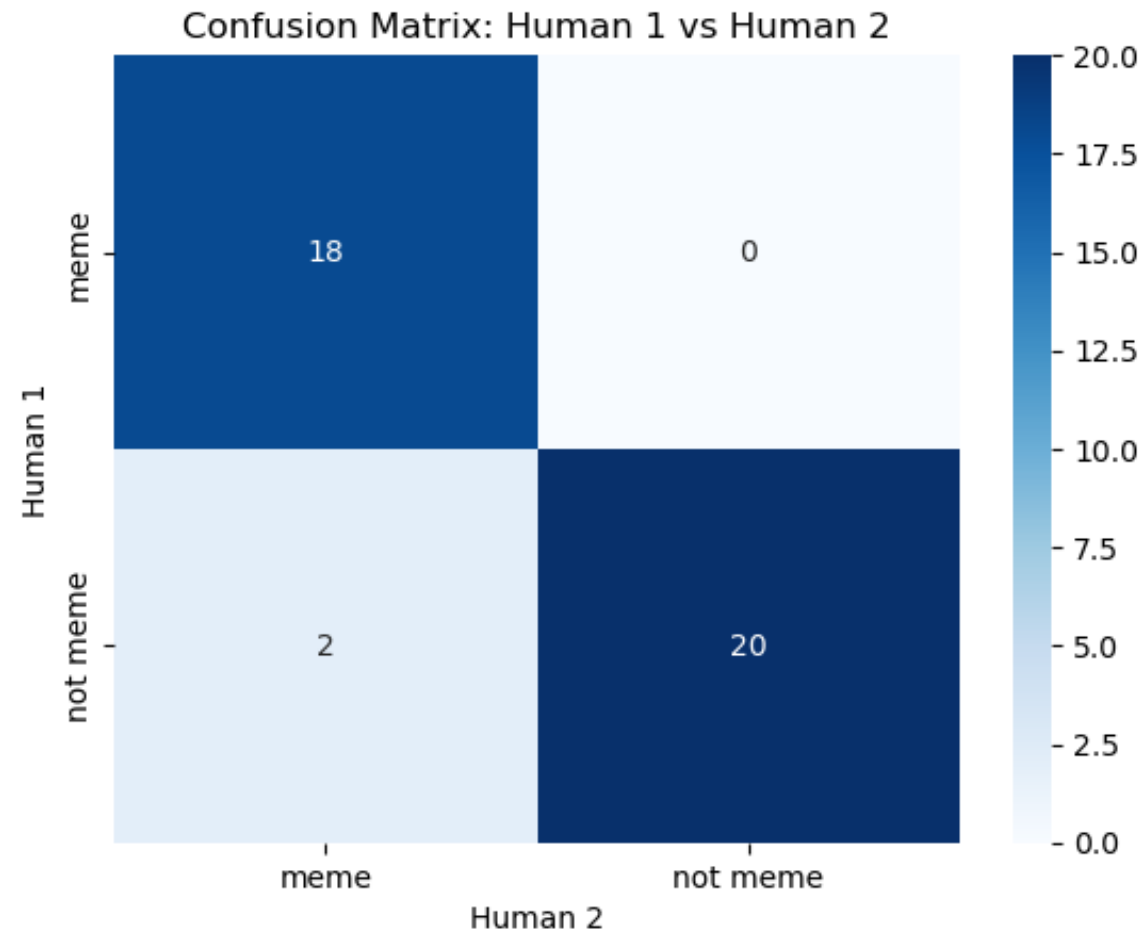


Model Comparison

Method	Annotations	Dice (meme)	Dice (not meme)	Avg.Dice
Method 1 (CNN)	Human 1	0.083333	0.607143	0.345238
Method 1 (CNN)	Human 2	0.076923	0.555556	0.316239
Method 2 (VLM)	Human 1	0.871795	0.878049	0.874922
Method 2 (VLM)	Human 2	0.878049	0.871795	0.874922



Human vs Human



Sentiment Analysis

We use the pre-trained and fined tuned model: cardiffnlp/twitter-roberta-base-sentiment (Barbieri et al.)

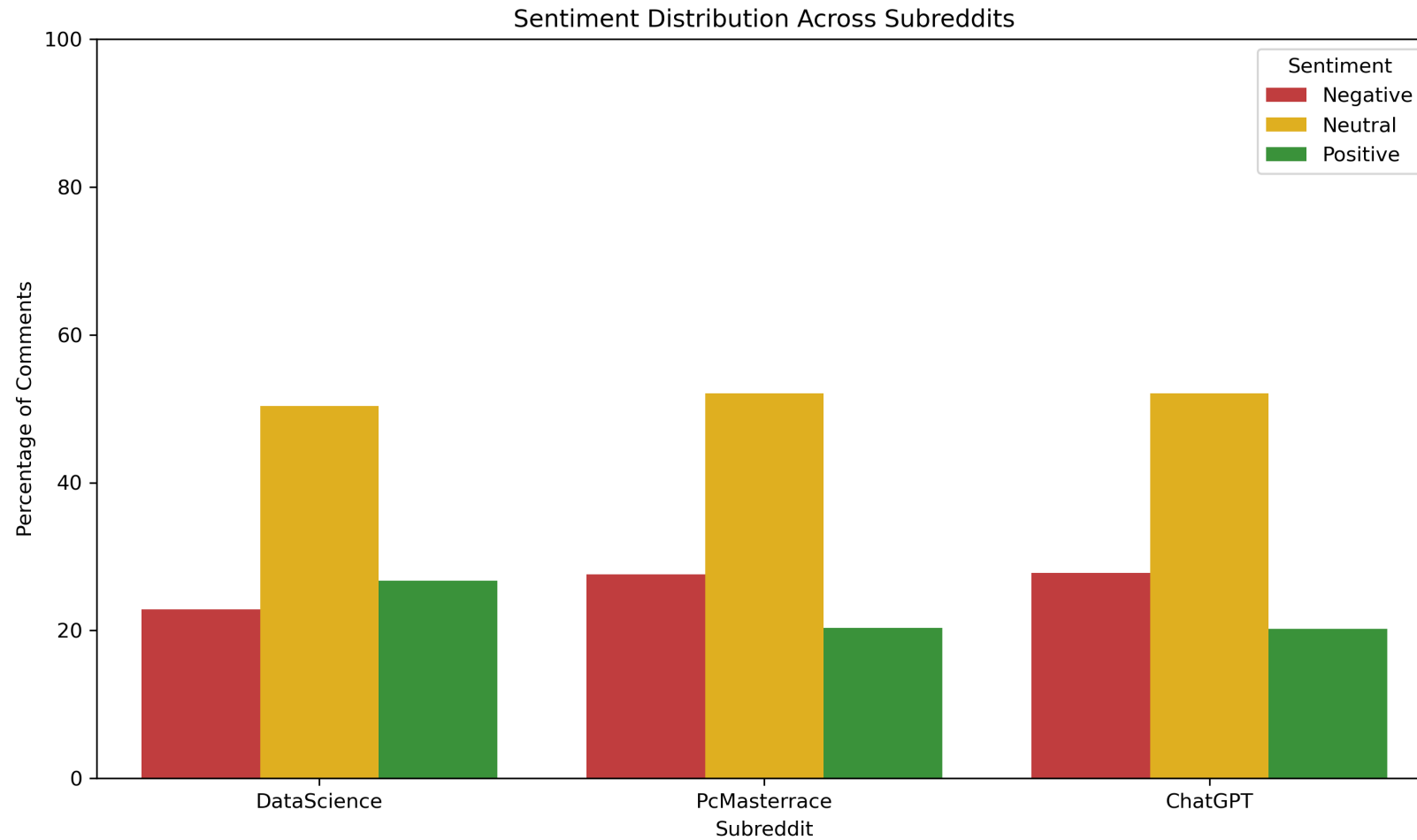
- Transformer (RoBERTa-base)
- Fine-tuned on English tweet data
- Labels: Positive, Neutral, Negative

We randomly sample a few comments and check quality of annotation.

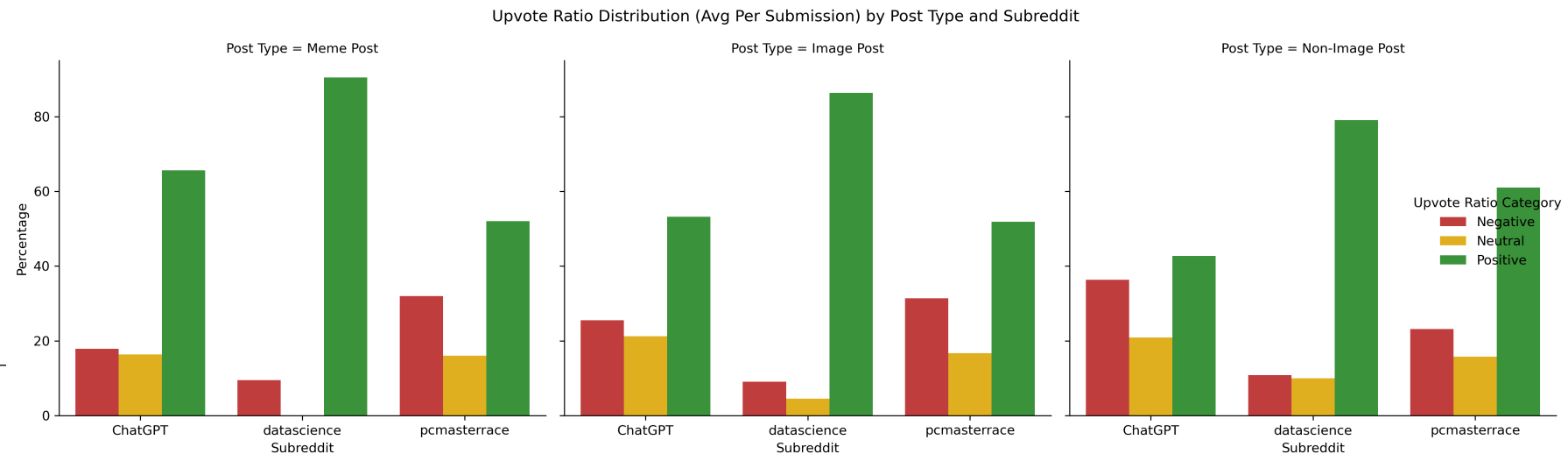
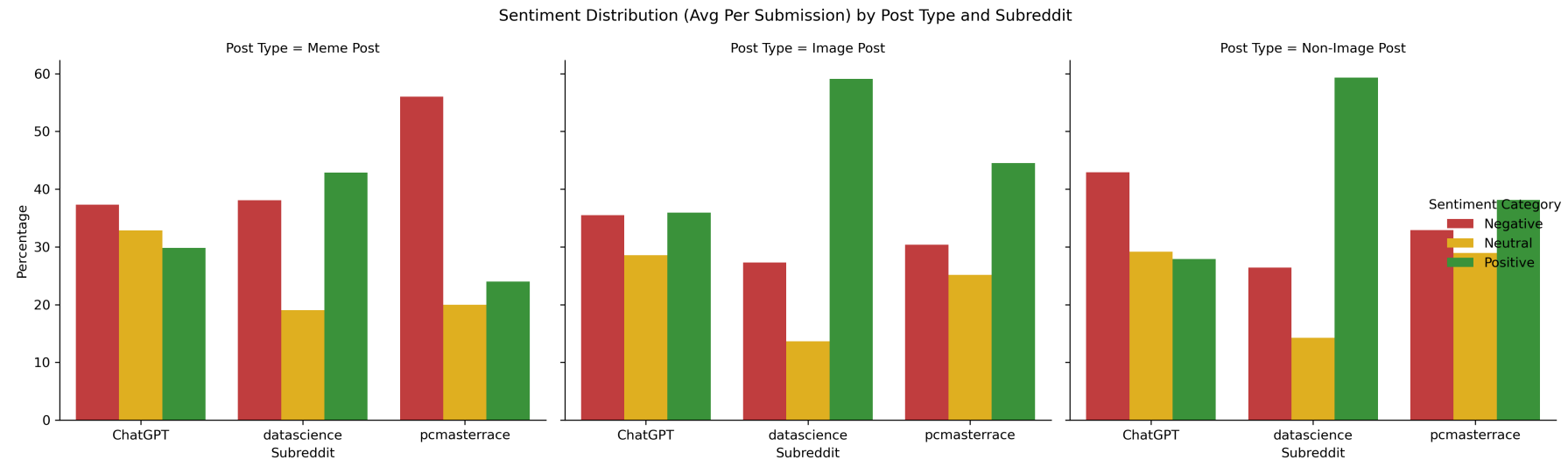
Filtering out Bots in the comments and Gifs.

Comment: I'm dead 📴
Sentiment: -1

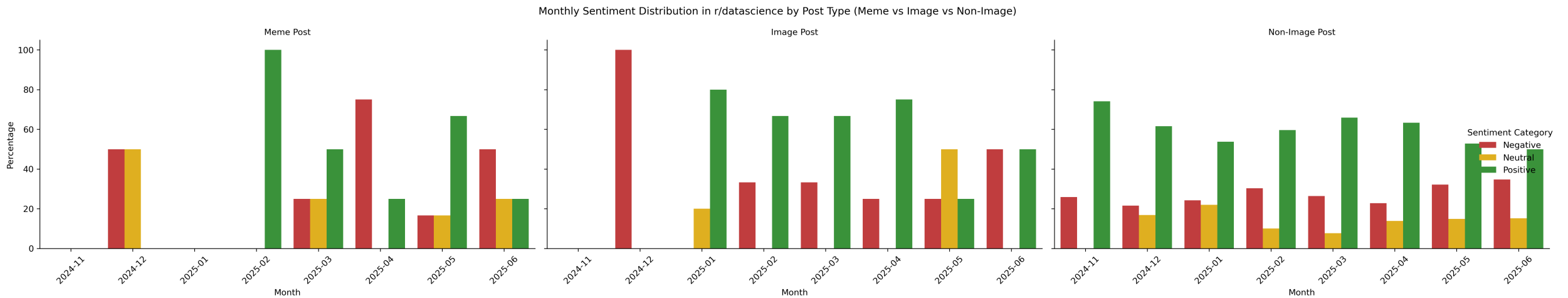
Results (Sentiment)



Results (Sentiment)



Results (Sentiment)



Social Network Analysis

We construct a social network – users directly interacting through comments

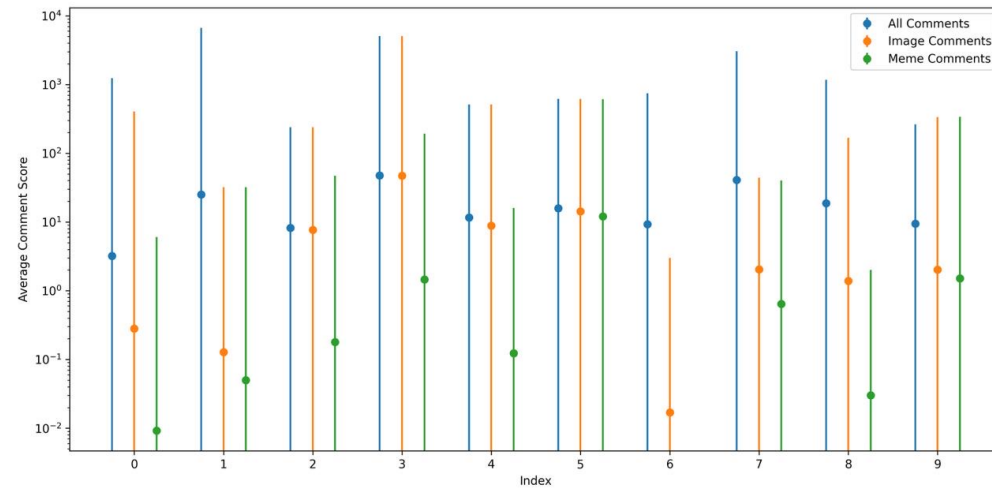
/r/datascience: N=8278 M=17190

/r/pcmasterrace: N=7788 M=10184

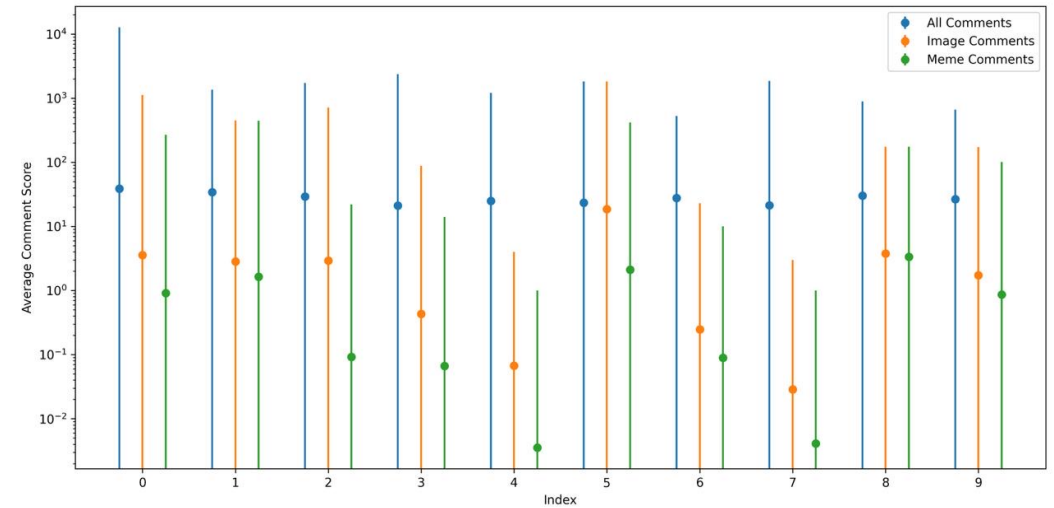
/r/ChatGPT: N=9657 M=12653

Social Network Analysis –Comment Score

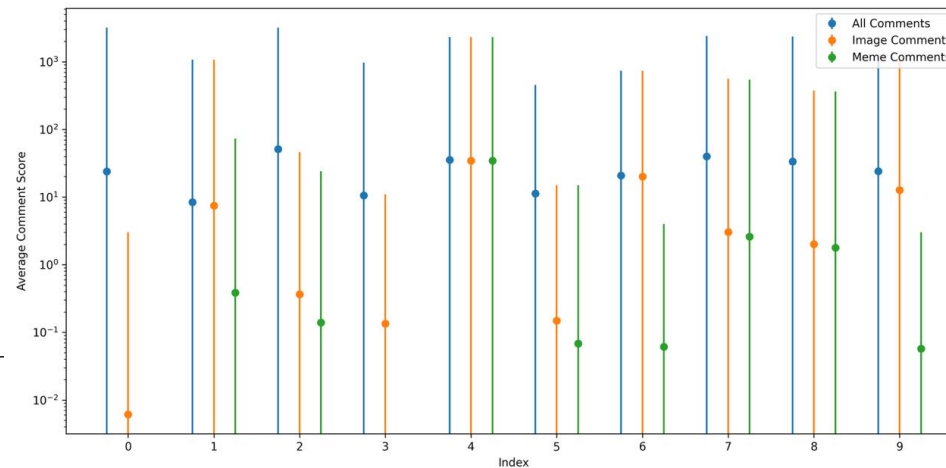
ChatGPT



Datascience



PCMasterRace



Limitations & Possible Extensions

1. Data Completeness and Restrictions of Reddit as a Platform
 2. Meme Classification based on specialized models
-
1. Include Sentiment in the Network Analysis
 2. Better Temporal Analysis for Impact on Discourse (Increase of Meme posts over time?)
 3. Add amount of engagement (Amount of comments and Upvotes) to the Analysis

